

Lesson 23 — Repeat and Keep Count

Matches LEGO Teacher Portal lesson: "Picking Petals".

Objective (I can...): I can use a repeat loop and keep a simple count of how many times it ran. **CS concept:** Repeat loops + collecting and presenting simple data (counting the repeats) · **Builds on:** Lesson 22 (loop bodies).

AI Connection (2 min, read aloud): AI keeps a tally as it practices — how many rounds it has done, how many it got right. *Ask the group:* "What would a robot count to know it practiced enough?"

Materials: LEGO CS & AI kit + device per pair, a big tally sheet or counting mat per pair. **Model reuse:** Loops model from Lesson 21 — code only.

60-minute flow - Warm-up (10): Tally clap — coach claps in a loop; kids make a tally mark each time and count the total. We can collect data on how many times something happened. - **Concept + demo (8):** Run a repeat-5 loop and show a counter on screen (or a light that flashes once per repeat); the number IS our data. Point to how the count grows. - **Build/Code (25):** Pairs code a loop that repeats an action and shows the count (screen number or flashes), then record the count on their tally sheet. Run with two different repeat numbers and write both down. *Wiggle break + swap at ~12 min.* - **Challenge (7):** Make a little bar of blocks (one brick per repeat) to "show" the count — a first picture graph. - **Share + tidy-up (10):** Pairs show their loop, read the count aloud, and hold up their brick-bar of the data. Model stays built.

Pair roles: Driver codes and runs the loop, Navigator tallies and reads the count. Swap at the wiggle break. (*Trio: third builds the brick-bar.*)

Differentiation: *Support* — count just one small loop. *Extension / fast-finisher* — compare counts from two loops and say which is bigger. *Reteach* — count each repeat out loud together and mark it as it happens.

Exit check: Pair runs a repeat loop and collects and shows a simple count of the repeats. **At-home (optional):** Count how many times you can hop in one loop of a song; write the number. ~5 min.
